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final technical notes on the "Chilean mummy" paper

These notes should now be okay. Please feel free to share, edit or have them reviewed by others?

Your friend, [REDACTED] in Sydney

*Repeat*

**Many "novel" mutations were found by PCR and sequencing, within the polymerase-copied DNA sample taken from a small mummy in Chile, that had been exposed to harsh local conditions in the desert there for 100 to 500 years: what might be the cause?**

The recovery of intact DNA from ancient samples is a subject fraught with technical difficulties. Most bone samples from even 100 years ago may show fragmentation to reduced mean sizes of 100-300 bp, plus oxidative damage to their bases which, if not eliminated from the sample before PCR, may appear as "misread bases" or "novel" genetic mutations in a final sequenced database.

The most common kind of damage involves deamination of C cytosine or A adenine bases, which later causes them to be read as U uracil or H hypoxanthine respectively, due to errors in base pairing when a polymerase enzyme copies the damaged base (see for example [https://www.mun.ca/biology/scarr/Nitrous\\_Acid\\_Mutagenesis.html](https://www.mun.ca/biology/scarr/Nitrous_Acid_Mutagenesis.html)). One standard method which may be used to reduce the frequency of misread C bases in a final sequenced product is to treat the aged or damaged sample with an enzyme UNG before PCR.

Pre-treatment with UNG: remove many C bases deaminated by environmental damage, yet still keep real genetic mutants which formed while the organism was alive.

Omit treatment with UNG: keep many C bases deaminated by environmental damage, and also keep real genetic mutants which formed while the organism was alive.

This enzyme uracil-N-glycosylase cuts a DNA strand at uracil U bases, and also at deaminated C bases which resemble uracil U (see <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4175022> or <https://academic.oup.com/nar/article/29/23/4793/2359260>).



How damaged by age was the Chilean mummy DNA sample? Could low-level environmental damage be something we need to worry about, especially if we plan to study long nuclear coding regions, and base our bioinformatics strategy around whether single point-mutations are seen in long coding regions of 500 to 2000 bp?

The authors write (edited slightly for clarity): *“The Ata DNA (sample) is relatively free of damage. The average DNA fragment size for Ata is 300 bp, consistent with a sample younger than 500 years.”*

Yet fresh undamaged DNA from a modern sample is much longer than that! We can therefore know for sure that its sugar-phosphate chains were seriously damaged by age. In other words, we cannot rely on any sugar-phosphate chain from that mummy sample being intact and undamaged beyond a mean length of 300 bp in double-stranded form. There might also be “nicks” within the single-stranded form of that DNA sample with an even smaller range of sizes (not mentioned in the published paper).

DNA bases such as A, G, C and T are nearly as susceptible to chemical and/or environmental damage as are the sugar-phosphate chains. Might some of those DNA bases have been damaged also in the mummy sample, at a mean spacing along any strand of 300 bases or less? How did the authors look for chemical damage to the DNA bases, which might give a false overestimate of “mutations” later in high-throughput sequencing, due to polymerase misreading of damaged bases? How did they try to correct for this problem, which is seen in almost all samples of ancient DNA older than 100 years, which have not been frozen away somewhere in a freezer or in the Arctic permafrost? (see for example <http://www.genetics.org/content/172/2/733>)

In Figure S1, they assess the extent of DNA damage by studying variations in sequence across small sequence-reads of just 101 bp. This method would only detect base-pair errors made **during sequencing** in vitro, mainly at the two ends of any small 101 bp fragment, due to polymerase misreading there.

*“Preserved DNA extracts may exhibit DNA damage or contaminants. We characterized the extent and type of DNA damage present by measuring nucleotide mis-incorporations, particularly cytosine deamination at the ends of fragments. We observed a very small increase in the frequency of C→T and G→A substitutions resulting from cytosine deamination at the 5' and 3' ends, respectively, with an approximately twofold difference in substitution frequency at the ends of the read versus the center.”*

In other words, this particular method would **not** be capable of detecting a low level of age-related environmental damage of bases, all across the original mummy DNA fragments of mean size 300 bp. Their final derived sequences were just assumed to be “real” and full of “novel” mutations, even if they were slightly different in sequence from

a human consensus. Once again, the mean size of DNA in their moderately-degraded sample was just 300 bp (as a double helix), and it was 100-500 years old. There is no possibility whatsoever that **some** of its bases did not suffer serious chemical damage after the mummy died, and before its genomic DNA was extracted, perhaps at a low level of 0.1-0.5% as suggested by its reduced DNA lengths?

Another test using mitochondrial DNA was shown in Fig. S2, yet was not described there in enough detail to be useful, for assessing the prevalence of very low-level mutations due to DNA damage. Thus the rates of nucleotide reading-error in nuclear genes, which might produce false and "novel" mutations by PCR and sequencing, and then affect later bioinformatics studies of long nuclear coding regions, would be expected to be only 0.1-0.5% or once every 200 to 1000 bp:

*"Overall we did not observe **much** damage in Ata's genome. As a result, the UNG treatment prior to amplification was **not** recommended (used). We also estimated contamination using the rate of mitochondrial heterozygosity. This analysis predicted the probability of authenticity of the specimen to be ~1.00, due to **limited** or no heterozygosity of the mitochondria."*

Since they knew that their mummy sample was moderately degraded to a mean length of 300 bp in double-stranded form, and they planned to study long 500-2000 bp nuclear coding-regions later, within which the presence of even **one** "false mutation" might seriously impair the bioinformatics analysis, why did they choose **not** to use a standard enzyme UNG, at least as a control, to remove most of the deaminated C bases prior to PCR?

This is likewise why studies of nuclear genes from ancient samples have been so problematic. It is easy to make some kind of "evolutionary tree" using sequences which are not quite accurate, without much effect on overall conclusions. Yet if we try to use DNA sequences of less than perfect quality to study long nuclear coding regions, as we might do when studying "fresh modern DNA" from medical patients, the entire exercise becomes fraught with technical difficulties. How do we know that what we are seeing was really there, while the organism was still alive, and not produced as an artefact after the organism died, then laid out in the Chilean desert for 100-500 years?

One way to check a single genomic database for accuracy would be to compare its derived sequences to many real mutations of long nuclear genes which we can see, in other much-larger databases of a similar kind, derived from the consensus of thousands of modern, human individuals. Do the specific locations of certain critical "mutations" which we see, for an ancient DNA sample, match the detailed locations of real genetic mutations which have been established through studying modern humans who might have given fresh blood or skin samples?



In fact, the investigators of that small Chilean mummy quickly saw lots of “novel” mutations in their final sequenced database, many of which did not match any other known mutations in current large, combined human databases. And so the “alarm bells” should have been ringing in their heads! What is really going on here? Are our DNA sequence data real, or could we have made a subtle mistake?

**Title: “Whole-genome sequencing of Atacama skeleton shows novel mutations linked with dysplasia”**

They found many “novel” DNA mutations, to which they ascribed critical and real genetic significances. Such a result is not impossible, but it needs to be checked very carefully, over and over again by a variety of methods and decisive controls, before it can be accepted as “fact”. That is how experimental science should be done.

The authors to their credit did “re-sequencing” of certain targeted regions, which was difficult because the mean size of DNA in that mummy sample was only 300 bp (their Table S5). Yet it was all done accordingly to the same methods as used before, and just confirmed that sequences from their whole-genome study were self-consistent and repeatable, rather than confirming that such DNA base changes were made while the mummy was still alive.

The burden of proof therefore seems to lie on those investigators (or others) to independently reproduce the occurrence of so many “novel” base mutations in the mummy sample (after PCR and sequencing), by repeating the same procedure, except this time using UNG or a similar enzyme, or other established methods to remove deaminated C bases (or other lesions) beforehand, then ask whether so many “novel” mutations still appear after sequencing, in the same places as before?

A full genomic-sequence analysis might be prohibitively expensive for every possible “control” set of conditions. Yet it would be relatively easy to check 6-12 “novel” mutations by PCR and sequencing, after applying various treatments which might eliminate damaged DNA bases, then ask whether the same “novel” mutations are still observed in the same places? Why were such important controls not asked for by the editor or referees of *Genome Research*?

In summary, how do we know that such “novel” mutations are real, and due to genetic changes while that small mummy was still alive, instead of environmental damage after the organism died, and was exposed to harsh conditions for >100 years in the Chilean desert? We seemingly do not.

Bioinformatics: how did they reach their unusual medical conclusion of a new “alien baby syndrome”?



Next, how did these authors come to their conclusion that a combination of known and “novel” mutations might be responsible for a strange “alien” body shape of that small mummy? Their explanations do not seem entirely convincing (paraphrased here for clarity):

*“We identified 64 coding region SNVs **predicted** to be deleterious. Next we found that the **majority** were **bone-associated**, such as ‘proportionate short stature’ and ‘11 pairs of ribs’. As a negative control, we ran similar analyses on a randomly-selected Peruvian female. There were **no** overlapping genes with mutations identified from the Ata genome present in this individual. The SNVs that we identified are **novel**, but (other) previously identified mutations in all seven genes are implicated in osteo-chondro-dysplasias, and represent plausible causes of Ata’s abnormal skull morphology, small stature, 10 ribs, and premature bone age.”*

Only 64 coding-region mutations out of millions were first predicted to be “deleterious”. Yet any single mutation in a coding region which changes an amino-acid can be “deleterious”, for example in sickle-cell anaemia.

Next a majority of those “chosen” 64 coding-region mutations were found to be located in “bone-associated genes”. While a modern Peruvian woman shows none there (0 of 64)? And such “novel” mutations were not found in current human databases?

If those were real “genetic mutations”, made while the mummy was alive, this result would require some kind of chemical mutagen which miraculously “targets” 7 bone-associated genes among 30,000 genes elsewhere in the genome. What might they say to further explain such unprecedented findings?

*“A **chance** combination of multiple known mutations and **novel** SNVs as **identified here** may explain Ata’s small stature, inappropriate rib count, abnormal cranial features, and perceived advanced bone age. The specimen was found in La Noria, one of the Atacama Desert’s many nitrate mining towns, which suggests a possible role for prenatal **nitrate exposure** leading to DNA damage.”*

Now they say that multiple mutations were not apparently targeted to 7 bone-associated genes, as suggested before, but were due purely to random chance. How can we reconcile this logic with their earlier bioinformatics conclusions?

Sodium nitrate as a cause of extensive genetic mutations in pregnant women, which causes the foetus to look like a small alien? This seems very unlikely (see for example <https://link.springer.com/article/10.1007/BF03174993>)

Three possible hypotheses: age-related artefacts, an entirely new medical disease, or past interbreeding with an unknown humanoid?



In order for the putative results of this study to be true, the authors would have had to identified an entirely new disease-causing syndrome, not previously known to medical science, which mysteriously (and for no known reason) caused an entire linked collection of bone-development genes within a human foetus to all mutate at the same time, thereby causing a normal Chilean mother to have an aborted baby which "looks like an alien". This concept seems somewhat akin to what happens when a normal man turns into a bone-mutated monster in the "Incredible Hulk" comic books, as the result of being exposed to gamma radiation in laboratory experiments.

It is not impossible that they have made such a great, Nobel-Prize-winning discovery. "Prions" and "quasicrystals" for example were doubted for many years, yet later shown to be true. We will be the first to congratulate them, if this "discovery" is later verified by other known cases? Say if this discovery were repeated in the case of a modern mother and her baby, from which fresh undamaged DNA could be obtained, that would be evidence in their favour. It might even be called **"alien baby syndrome"**.

Still respected sceptics such as Carl Sagan have said that "extraordinary claims require extraordinary evidence". Here we see a case of "extraordinary claims" which have been based on weak or doubtful evidence, and the same sceptical logic has not been applied.

Once all of those "novel" mutations from an aged and desert-exposed mummy sample are tested for factual genetic reality, using a series of established controls, then we might be able to say more. We need to see more controls for possible age-related artefacts, using the same polymerase as used in high-throughput sequencing, yet with various pre-treatments to remove damaged bases (UNG, heat, etc.) before we can be sure that any of those "novel" mutations are real, and certainly before we try to ascribe any possible causes to them (for example exposure to nitrate, past hybridization, etc.)

If this were any other study of an aged mummy from mainstream "DNA archaeology", it is hard to see how the paper could have been accepted in its current form. Out of the thousands of ancient mummies or bone samples studied so far, none have ever shown an entire database filled with "novel" but real genetic mutations. Again it is not impossible that such mutations could be real, because this is such a strange-looking sample. Yet to attribute all such mutations to "normal human biology" or "nitrate mutagenesis" seems unlikely. Perhaps something else may be going on here?

Other workers from DNA archaeology have been reluctant to draw firm conclusions concerning long nuclear genes for this very reason. Their comments on this paper would be welcome (see for example <http://www.mdpi.com/2073-4425/9/3/135> or <https://www.forensicmag.com/article/2016/03/nuclear-dna-analyzed-400000-year-old-bones>), even though they have not yet uncovered any new medical syndromes, but are just studying multiple-human-species ancestries.

If any of the mutations reported in this small Chilean mummy matched for example Neanderthal or Denisovan nuclear genes, that would be of great interest, since some of those pseudo-human species apparently interbred with modern humans long ago (see <https://en.wikipedia.org/wiki/Denisovan> or <https://en.wikipedia.org/wiki/Neanderthal>). Might the past existence of several pseudo-human species imply that some other “unknown” humanoid species interbred with that Chilean woman, hundreds of years ago, although their hybrid offspring may not have been fully viable?

All of these three hypotheses: (1) age-related artefacts, (2) new medical syndrome, or (3) past interbreeding, might usefully be considered as we move forward.

THE END